BEYOND DEVICE INVARIANCE: SCENE-CONDITIONED DEVICE FLOW MODELING FOR CROSS-DEVICE ACOUSTIC SCENE CLASSIFICATION

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ABSTRACT

Acoustic scene classification (ASC) often suffers from device mismatch. Many existing methods mitigate this mismatch by making scene recognition less sensitive to device effects. While effective, this principle implicitly assumes that device effects can be mitigated without compromising scene-discriminative information. However, this assumption may not always hold: recording devices can alter how scene-relevant acoustic cues are observed, entangling device-conditioned variation with scene-discriminative structure in the representation space. Consequently, reducing sensitivity to device effects may also weaken scene-discriminative information.

Motivated by this observation, we propose AnchorFlow, a plug-and-play training mechanism for scene-conditioned device flow modeling. AnchorFlow represents device-induced variation within each acoustic scene using distributional anchors and closed-form device flows, treating such variation as structured training information rather than solely as a nuisance factor. Experiments across multiple ASC datasets and backbones show consistent improvements in cross-device generalization, with particularly strong gains on unseen devices and no additional inference overhead. Code is available at `github.com/manifold112/AnchorFlow_ICASSP2027`.

***Index Terms*—** Acoustic scene classification, cross-device generalization, device mismatch, representation learning

1. INTRODUCTION

Acoustic scene classification (ASC) aims to recognize acoustic environments from audio recordings [1, 2]. In practice, differences in microphone frequency responses across recording devices can alter scene-relevant acoustic characteristics, causing models trained on a limited set of devices to rely on device-correlated patterns that do not generalize well to unseen devices [3, 4]. Consequently, robustness to device mismatch has become a central problem in ASC [5–8].

To mitigate device mismatch, existing methods intervene at the observation or representation level [9]. At the observation level, prior work compensates for device-dependent spectral responses through spectrum correction [10], simulates recording-condition variation using impulse-response augmentation [11, 12], or diversifies frequency-response characteristics through augmentation [13–15]. At the representation level, robustness is pursued by normalizing device-sensitive feature statistics [16], aligning feature distributions across devices [17–19], or suppressing device-specific information while preserving scene-discriminative cues [20]. Despite these different mechanisms, these approaches aim to reduce the influence of device-induced variation on scene recognition.

While effective, this view implicitly assumes that device-conditioned variation is sufficiently separable from scene-discriminative information, such that reducing device effects does not compromise the acoustic cues required for scene recognition. However, recording devices can alter how scene-relevant acoustic cues are captured and represented [21–23], with their effects potentially interacting with different scene-relevant cues across scenes, as shown in Fig. 1. Enforcing domain invariance can discard predictive information when domain and label structure are not fully separable [24]. We therefore hypothesize that explicitly modeling scene-conditioned device variation can improve generalization to unseen recording conditions.

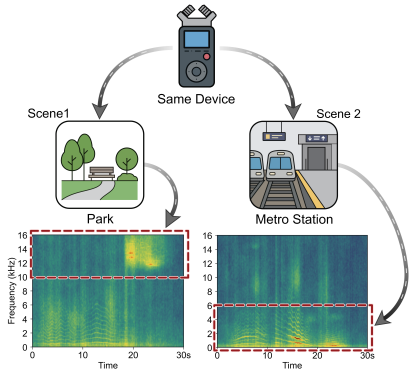


Fig. 1. Scene-conditioned device effects in ASC. A device change affects scene-relevant cues differently across scenes.

To examine this hypothesis, we propose AnchorFlow, a plug-and-play training mechanism for scene-conditioned device flow modeling. AnchorFlow characterizes device variation within each acoustic scene using distributional anchors and constructs closed-form flows between same-scene device conditions. These flows describe structured trajectories of how scene representations vary across recording devices while preserving sample-specific characteristics. Rather than enforcing device invariance, AnchorFlow leverages these trajectories to maintain scene-discriminative behavior across device conditions. Device labels are used only during training, and the original inference pipeline remains unchanged.

We evaluate AnchorFlow on two DCASE cross-device ASC protocols and a CochlScene evaluation with microphone responses from MicIRP. Across these settings and backbones, AnchorFlow consistently improves cross-device generalization, with average gains of 3.31 percentage points on unseen devices and 3.22 points on held-out microphone responses. AnchorFlow operates exclusively during training and leaves the original inference pipeline unchanged, enabling straightforward integration into existing ASC systems.

2. OUR PROPOSED ANCHORFLOW

Let $\mathcal{D}_{\text{tr}} = \{(\mathbf{x}_i, y_i, d_i)\}_{i=1}^{N}$ denote the training set, where $y_i \in \mathcal{Y}$ and $d_i \in \mathcal{D}_{\text{obs}}$ are the scene and observed-device labels. For an

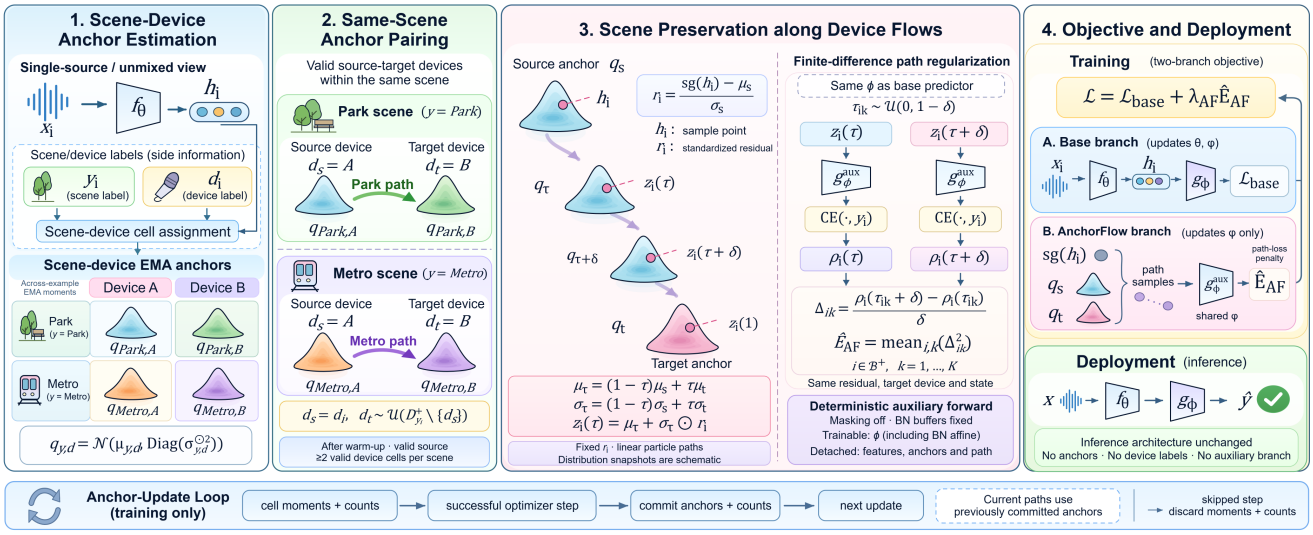


Fig. 2. Overview of AnchorFlow. Scene–device distributional anchors are paired within each acoustic scene to construct closed-form device flows. Residual-preserving trajectories are then used to preserve scene-discriminative behavior along modeled device transitions.

existing ASC model, we formulate

$$\mathbf{h}_i = f_\theta(\mathbf{x}_i), \qquad \mathbf{s}_i = g_\phi(\mathbf{h}_i), \tag{1}$$

where $\mathbf{h}_i \in \mathbb{R}^p$ is a selected representation and g_ϕ is the remaining scene predictor. We denote its original training objective by $\mathcal{L}_{\text{base}}(\theta, \phi)$. AnchorFlow is applied only during training and uses device labels to model scene-conditioned device variation, while the original inference model $g_\phi \circ f_\theta$ remains unchanged.

2.1. Scene-Conditioned Device Anchors

For each observed scene–device pair (y, d), we approximate the conditional representation distribution by

$$q_{y,d} = \mathcal{N}\big(\boldsymbol{\mu}_{y,d}, \mathrm{Diag}(\boldsymbol{\sigma}^{\odot 2}_{y,d})\big)\,. \tag{2}$$

Joint scene–device indexing allows device-conditioned location and scale to vary across acoustic scenes. We use a diagonal Gaussian as a lightweight approximation for scene–device cells, which avoids estimating high-dimensional full covariances and enables closed-form coordinate-wise transport in Sec. 2.2.

Let $\mathcal{I}_{y,d}$ denote the samples from scene–device cell (y, d) in an effective update, with $c_{y,d} = |\mathcal{I}_{y,d}|$. We compute the first and second detached raw moments as

$$\overline{\mathbf{h}^{\odot r}}_{y,d} = \frac{1}{c_{y,d}} \sum_{i \in \mathcal{I}_{y,d}} [\mathrm{sg}(\mathbf{h}_i)]^{\odot r}, \qquad r \in \{1, 2\}, \tag{3}$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient. For each nonempty cell, these moments update exponential moving averages

$$\begin{aligned} \mathbf{m}^{(r)}_{y,d} &\leftarrow \beta \mathbf{m}^{(r)}_{y,d} + (1-\beta)\overline{\mathbf{h}^{\odot r}}_{y,d}, \quad r \in \{1, 2\}, \\ \boldsymbol{\mu}_{y,d} &= \mathbf{m}^{(1)}_{y,d}, \\ \boldsymbol{\sigma}_{y,d} &= \Big[\max\Big(\mathbf{m}^{(2)}_{y,d} - \boldsymbol{\mu}^{\odot 2}_{y,d}, \nu_{\min}\Big) + \varepsilon\Big]^{1/2}, \end{aligned} \tag{4}$$

where β is the EMA decay, while $\nu_{\min} \geq 0$ and $\varepsilon > 0$ ensure a positive scale. Each cell is initialized on its first nonempty update, and current paths use anchors from the preceding update.

Let $\mathcal{D}^+_y$ denote the observed devices with scene-y anchors estimated from at least $n_{\min}$ samples. After warm-up, for a sample i with $d_i \in \mathcal{D}^+_{y_i}$ and $|\mathcal{D}^+_{y_i}| \geq 2$, we set $d_s = d_i$ and draw

$$d_t \sim \mathcal{U}\big(\mathcal{D}^+_{y_i} \setminus \{d_s\}\big)\,. \tag{5}$$

Thus, transitions are restricted to anchors within the same scene, so that the variation is attributed to device conditions rather than scene changes, and no paired recordings across devices are required.

2.2. Closed-Form Device Flows

For an eligible sample i, let $q_s = q_{y_i,d_s}$ and $q_t = q_{y_i,d_t}$ denote the source and target anchors, with statistics $(\boldsymbol{\mu}_s, \boldsymbol{\sigma}_s)$ and $(\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t)$, respectively. We define the standardized residual

$$\mathbf{r}_i = [\mathrm{sg}(\mathbf{h}_i) - \boldsymbol{\mu}_s] \oslash \boldsymbol{\sigma}_s, \tag{6}$$

where $\oslash$ denotes element-wise division. For $\tau \in [0, 1]$, we construct the device-flow trajectory as

$$\begin{aligned} \boldsymbol{\mu}_\tau &= (1-\tau)\boldsymbol{\mu}_s + \tau\boldsymbol{\mu}_t, \\ \boldsymbol{\sigma}_\tau &= (1-\tau)\boldsymbol{\sigma}_s + \tau\boldsymbol{\sigma}_t, \\ \mathbf{z}_i(\tau) &= \boldsymbol{\mu}_\tau + \boldsymbol{\sigma}_\tau \odot \mathbf{r}_i. \end{aligned} \tag{7}$$

The interpolated statistics define $q_\tau = \mathcal{N}(\boldsymbol{\mu}_\tau, \mathrm{Diag}(\boldsymbol{\sigma}^{\odot 2}_\tau))$, which traces the 2-Wasserstein geodesic between q_s and q_t for diagonal Gaussians [25]. Since $\mathbf{z}_i(0) = \mathrm{sg}(\mathbf{h}_i)$ and $\mathbf{z}_i(1) = \boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \mathbf{r}_i$, fixing $\mathbf{r}_i$ preserves the standardized residual along the interpolated anchors and yields a linear particle path.

Let $\Delta\boldsymbol{\mu} = \boldsymbol{\mu}_t - \boldsymbol{\mu}_s$ and $\Delta\boldsymbol{\sigma} = \boldsymbol{\sigma}_t - \boldsymbol{\sigma}_s$. The trajectory satisfies

$$\frac{d\mathbf{z}}{d\tau} = \mathbf{v}_\tau(\mathbf{z}), \qquad \mathbf{v}_\tau(\mathbf{z}) = \Delta\boldsymbol{\mu} + \Delta\boldsymbol{\sigma} \odot [(\mathbf{z} - \boldsymbol{\mu}_\tau) \oslash \boldsymbol{\sigma}_\tau]\,. \tag{8}$$

Along $\mathbf{z}_i(\tau)$, the velocity is constant, $\mathbf{v}_i = \Delta\boldsymbol{\mu} + \Delta\boldsymbol{\sigma} \odot \mathbf{r}_i$. Thus, the device transition admits an explicit flow field, with both the trajectory and velocity available in closed form, requiring neither a learned transport model nor numerical ODE integration.

Table 1. Results on DCASE ASC protocols over five runs (mean ± SD). AF-enhanced results are bold, and average rows are shaded.

DCASE 2020 Task 1A									
Backbone	Seen			Unseen			Overall		
	Base	+AF	Δ	Base	+AF	Δ	Base	+AF	Δ
CP-Mobile [26]	74.72±0.46	**76.04±0.43**	+1.32	66.57±0.57	**69.70±0.61**	+3.13	72.00±0.39	**73.93±0.36**	+1.93
BC-ResNet-ASC-8 [27]	72.09±0.48	**73.57±0.47**	+1.48	64.34±0.52	**67.78±0.55**	+3.44	69.51±0.39	**71.64±0.39**	+2.13
TF-SepNet-64 [28]	73.41±0.50	**75.02±0.42**	+1.61	64.14±0.52	**67.85±0.57**	+3.71	70.32±0.27	**72.63±0.31**	+2.31
PaSST-S [29]	76.34±0.46	**77.41±0.54**	+1.07	74.34±0.65	**77.02±0.60**	+2.68	75.67±0.51	**77.28±0.48**	+1.61
Average	74.14	**75.51**	+1.37	67.35	**70.59**	+3.24	71.88	**73.87**	+1.99

DCASE 2024 Task 1 (split100)									
Backbone	Seen			Unseen			Overall		
	Base	+AF	Δ	Base	+AF	Δ	Base	+AF	Δ
CP-Mobile [26]	64.40±0.44	**65.64±0.42**	+1.24	53.10±0.58	**56.37±0.55**	+3.27	60.63±0.47	**62.55±0.38**	+1.92
BC-ResNet-ASC-8 [27]	63.10±0.40	**64.47±0.45**	+1.37	52.00±0.50	**55.57±0.57**	+3.57	59.40±0.35	**61.50±0.34**	+2.10
TF-SepNet-64 [28]	64.80±0.38	**66.32±0.35**	+1.52	53.70±0.49	**57.62±0.58**	+3.92	61.10±0.24	**63.42±0.28**	+2.32
PaSST-S [29]	62.40±0.39	**63.36±0.36**	+0.96	55.90±0.48	**58.62±0.59**	+2.72	60.23±0.26	**61.78±0.25**	+1.55
Average	63.68	**64.95**	+1.27	53.68	**57.05**	+3.37	60.34	**62.31**	+1.97

2.3. Scene Preservation along Device Flows

To enforce scene preservation along device flows, we use a deterministic auxiliary evaluation g_ϕ^{aux} that shares the trainable parameters ϕ with g_ϕ. Stochastic masking is disabled and batch-normalization statistics are fixed during this auxiliary forward. We define the scene loss along a device-flow trajectory as

$$\rho_i(\tau; d_t) = \ell_{\mathrm{scn}}\big(g_\phi^{\mathrm{aux}}(\mathbf{z}_i(\tau)), y_i\big)\,, \tag{9}$$

where ℓ_{scn} denotes the cross-entropy scene loss. By the chain rule,

$$\rho_i'(\tau; d_t) = \nabla_{\mathbf{z}} \ell_{\mathrm{scn}}\big(g_\phi^{\mathrm{aux}}(\mathbf{z}), y_i\big)^\top \mathbf{v}_i \Big|_{\mathbf{z}=\mathbf{z}_i(\tau)}\,, \tag{10}$$

which is the directional derivative of the scene loss along the modeled device flow and therefore measures how rapidly scene-discriminative behavior changes under the device transition.

We therefore define the AnchorFlow energy as

$$\mathcal{E}_{\mathrm{AF}} = \mathbb{E}_{i,d_t}\left[\int_0^1 \big[\rho_i'(\tau; d_t)\big]^2\, d\tau\right]. \tag{11}$$

By the Cauchy–Schwarz inequality,

$$|\rho_i(1; d_t) - \rho_i(0; d_t)|^2 \le \int_0^1 \big[\rho_i'(\tau; d_t)\big]^2\, d\tau, \tag{12}$$

Thus, controlling the path energy limits scene-loss variation along the modeled transitions between observed device conditions.

2.4. Finite-Difference Path Regularization

Direct optimization of the continuous path energy requires higher-order differentiation. We instead use a finite-difference approximation with ordinary first-order backpropagation. Let $\mathcal{B}^+$ denote the eligible samples in an update. For each $i \in \mathcal{B}^+$, we draw K target devices and sample $\tau_{ik} \sim \mathcal{U}(0, 1-\delta)$, where $0 < \delta < 1$. We define

$$\begin{aligned} \Delta_{ik} &= \frac{\rho_i(\tau_{ik}+\delta; d_{t,ik}) - \rho_i(\tau_{ik}; d_{t,ik})}{\delta}, \\ \widehat{\mathcal{E}}_{\mathrm{AF}} &= \frac{1}{|\mathcal{B}^+|K} \sum_{i\in\mathcal{B}^+} \sum_{k=1}^{K} \Delta_{ik}^2, \\ \mathcal{L} &= \mathcal{L}_{\mathrm{base}} + \lambda_{\mathrm{AF}} \widehat{\mathcal{E}}_{\mathrm{AF}}. \end{aligned} \tag{13}$$

Since Δ_{ik} estimates $\rho_i'(\tau_{ik}; d_{t,ik})$ by forward finite differences, $\widehat{\mathcal{E}}_{\mathrm{AF}}$ provides a stochastic surrogate for the continuous path energy in Eq. (11), approaching it in expectation as $\delta \to 0$. The two loss evaluations in each Δ_{ik} share the same residual, target device, and auxiliary predictor state, so their difference reflects movement along the same device flow. If $\mathcal{B}^+ = \varnothing$, we set $\widehat{\mathcal{E}}_{\mathrm{AF}} = 0$.

Features, anchors, and flow trajectories are detached from the AnchorFlow branch, while g_ϕ^{aux} remains differentiable. Therefore,

$$\begin{aligned} \nabla_\theta \mathcal{L} &= \nabla_\theta \mathcal{L}_{\mathrm{base}}, \\ \nabla_\phi \mathcal{L} &= \nabla_\phi \mathcal{L}_{\mathrm{base}} + \lambda_{\mathrm{AF}} \nabla_\phi \widehat{\mathcal{E}}_{\mathrm{AF}}. \end{aligned} \tag{14}$$

This detachment prevents the auxiliary objective from reshaping the representation to reduce the path loss. At inference, the auxiliary branch is removed and $g_\phi \circ f_\theta$ remains unchanged.

3. EXPERIMENT

3.1. Experimental Setup

Evaluation protocols. We evaluate AnchorFlow on two standard cross-device ASC benchmarks: DCASE 2020 Task 1A [30] using TAU 2020 Mobile (TAU20) [31], and DCASE 2024 Task 1 split100 [5] using TAU 2022 Mobile (TAU22) [32]. In both protocols, devices A–C and S1–S3 are observed during training, while S4–S6 are held out for unseen-device evaluation. TAU20 evaluates 10-s recordings by averaging predictions over ten 1-s segments, whereas TAU22 evaluates individual 1-s segments under the official protocol. For additional external validation beyond the DCASE settings, we use the user-disjoint CochlScene partitions [33] with nine MicIRP microphone responses [34], six observed and three held out.

Backbones and training. We use CP-Mobile [26], BC-ResNet [27], TF-SepNet [28], and PaSST-S [29]. For each backbone, matched baseline–AnchorFlow pairs use an identical training recipe, including the optimizer, learning-rate schedule, number of epochs, batch size, data preprocessing, and augmentation, with the same five random seeds. AnchorFlow is the only additional training component, and results are reported as mean ± standard deviation.

For the controlled comparisons in Sec. 3.3, all methods use the same TF-SepNet-64 backbone and training recipe, with method-specific operations added only as required. All method-specific hyperparameters are selected using source-only information, with

no held-out-device information used for training, hyperparameter selection, or model selection. We set $(\lambda_{\mathrm{AF}}, \delta, K) = (0.3, 0.20, 1)$. Anchor parameters are $\beta = 0.99$, $n_{\min} = 32$, $\nu_{\min} = 10^{-6}$, and $\varepsilon = 10^{-5}$, with a one-epoch warm-up.

3.2. Main Results

Table 1 shows that AnchorFlow consistently improves all four backbones under both DCASE protocols. The gains are systematically larger on unseen than on seen devices, indicating that the improvement is primarily associated with stronger cross-device generalization rather than better fitting the observed conditions. The consistent trend across diverse backbones further suggests that AnchorFlow is not tied to a particular model architecture.

3.3. Compatibility and Ablation Analysis

Table 2(a) compares AnchorFlow with representative device-robust methods under the same TF-SepNet-64 backbone. All methods use only devices A–C and S1–S3 during training, with no information from S4–S6 used for training or model selection.

Among standalone methods, AnchorFlow achieves the highest overall accuracy on both protocols, while RFN and Freq-MS perform better on the unseen-device metric. This indicates a more balanced improvement across seen and unseen conditions. Combining AnchorFlow with RFN, Freq-MS, or DIR further improves performance, showing that device-flow modeling is compatible with both normalization and device-oriented augmentation.

Table 2(b) examines the main design choices of AnchorFlow. The full model consistently outperforms all ablated variants. Mean-only transport is the strongest ablation, while removing scene conditioning, replacing the structured flow with matched isotropic perturbations, or using only endpoint preservation further reduces performance. The non-detached variant falls below the baseline, supporting the use of detached device-flow geometry.

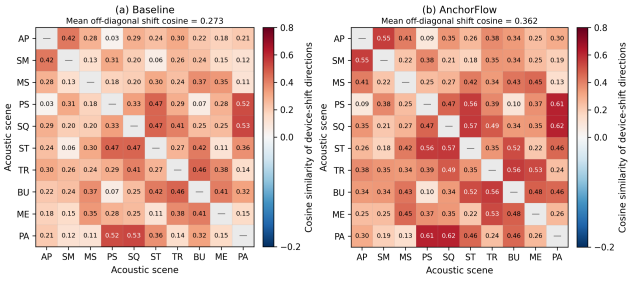


Fig. 3. Cross-scene cosine similarity of device-shift directions for (a) the baseline and (b) AnchorFlow. Each entry denotes the cosine similarity between the device-shift directions of a scene pair, with mean off-diagonal similarities of 0.273 and 0.362, respectively.

3.4. Mechanism Analysis and External Validation

Scene–device geometry. We fit an additive scene-plus-device model to scene–device mean embeddings and use the unexplained interaction ratio (IR) to quantify variation that cannot be explained by independent scene and device effects. A within-scene device-label permutation test provides the null reference. Table 3(a) shows significant non-additive scene–device interaction for both the baseline and AnchorFlow ($p < 0.005$), supporting the presence of scene-dependent device effects.

Table 2. Controlled comparisons and component ablations with TF-SepNet-64. Best means are bold, and AnchorFlow is shaded.

(a) Comparisons and compatibility

Method	DCASE 2020 Task 1A		DCASE 2024 Task 1 (split100)	
	Overall	Unseen	Overall	Unseen
Baseline	70.32±0.27	64.14±0.52	61.10±0.24	53.70±0.49
UADA [18]	70.69±0.38	66.87±0.52	61.57±0.10	56.51±0.48
FBS+SDA [19]	71.31±0.40	67.54±0.58	62.13±0.36	57.34±0.56
RFN [16]	71.77±0.32	68.99±0.55	62.77±0.28	58.90±0.48
Freq-MS [15]	71.97±0.31	68.28±0.51	63.30±0.32	59.49±0.48
DIR [11]	71.76±0.34	67.61±0.56	62.84±0.31	58.43±0.54
ADIR [12]	71.91±0.35	68.04±0.58	63.06±0.32	59.05±0.55
AnchorFlow	72.63±0.31	67.85±0.57	63.42±0.28	57.62±0.58
DIR + Freq-MS	72.19±0.36	69.31±0.61	63.52±0.34	60.56±0.57
AF + RFN	73.00±0.39	**70.24±0.69**	63.78±0.34	60.62±0.62
AF + Freq-MS	**73.16±0.38**	70.12±0.66	**64.02±0.35**	**61.16±0.64**
AF + DIR	73.08±0.40	69.82±0.67	63.88±0.34	60.54±0.61

(b) Component ablations on DCASE 2024

Variant	Seen	Unseen	Overall
Baseline	64.80±0.38	53.70±0.49	61.10±0.24
Scene-agnostic	65.64±0.39	55.78±0.53	62.35±0.28
Mean-only	65.94±0.44	56.64±0.59	62.84±0.30
Endpoint	65.86±0.46	56.44±0.57	62.72±0.32
Non-detached	63.55±0.32	51.95±0.46	59.68±0.27
Matched isotropic	65.56±0.43	55.66±0.54	62.26±0.31
AnchorFlow	**66.32±0.35**	**57.62±0.58**	**63.42±0.28**

Table 3. Mechanism analysis on TAU22 and external validation on CochlScene. AnchorFlow is shaded.

(a) Scene–device interaction on TAU22

Method	IR	p-value
Baseline	0.01956±0.00072	< 0.005
AnchorFlow	0.02094±0.00079	< 0.005

(b) External validation on CochlScene

Method	Seen	Unseen	Overall
Baseline	76.13±0.35	65.22±0.60	72.49±0.40
AnchorFlow	**77.96±0.39**	**68.44±0.56**	**74.79±0.38**

Fig. 3 further examines the geometry of these effects through the cross-scene cosine similarity of device-shift directions. AnchorFlow increases the mean off-diagonal similarity from 0.273 to 0.362, showing that device-induced shifts become more coherent across scenes. Yet several scene pairs remain weakly aligned, suggesting that AnchorFlow organizes scene-conditioned device variation rather than collapsing it into a scene-independent shift.

External response shift. We further evaluate AnchorFlow on the user-disjoint CochlScene [33] setting with MicIRP microphone responses [34]. As shown in Table 3(b), AnchorFlow improves performance under both seen and held-out microphone responses, with a larger gain under unseen responses. This consistent advantage under a distinct dataset and response-shift setting supports generalization beyond the DCASE device protocols.

4. CONCLUSION

We presented AnchorFlow for cross-device acoustic scene classification, which models device variation as scene-conditioned structure rather than solely as a nuisance factor. Across four backbones and two DCASE protocols, AnchorFlow consistently improves cross-device generalization, with larger gains on unseen devices. Its compatibility with existing device-robust methods and improvements under external microphone-response shifts further support the generality of the proposed modeling strategy. AnchorFlow operates only during training and leaves the inference pipeline unchanged.

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